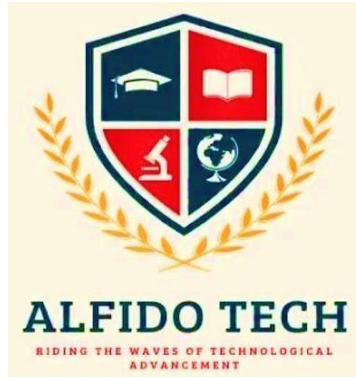




Random User Directory

Frontend Development Internship – Task 3 Report

API Integration Project **Academic year: 2026-2027**

Submitted To: Alfido Tech

Internship Domain: Frontend Development

Task Number: Task 3 – API integration

Project Title: Random user directory

Submitted By: NIREEKSHA M R

Duration: 30 Days

Technology Used: HTML, CSS, JavaScript

INDEX

Sl. No. Contents

1	Abstract
2	Objective
3	Introduction
4	Technologies Used
5	Project Description
6	Features
7	API Used
8	Folder Structure
9	Methodology / Implementation
10	Learning Outcomes
11	Conclusion
12	GitHub profile link

1. Abstract

The **Random User Directory** is a frontend web application developed to demonstrate the integration of a public REST API using JavaScript. The application retrieves user information from the Random User API and dynamically displays it in an attractive and responsive card-based interface.

The project focuses on real-world frontend development concepts such as asynchronous programming, API integration, DOM manipulation, event handling, and responsive web design. It also includes useful features such as a search bar for filtering users, a reload button to fetch new user data, a loading indicator during data retrieval, and error handling to improve the user experience.

This project provided practical experience in building interactive web applications that communicate with external services while maintaining a clean and user-friendly interface.

2. Objective

The primary objective of this project is to design and develop a responsive web application that integrates a public REST API to fetch and display user information dynamically.

The project aims to:

- Understand REST API integration.
 - Learn asynchronous programming using JavaScript.
 - Fetch real-time data from external sources.
 - Display dynamic content using DOM manipulation.
 - Develop responsive user interfaces.
 - Implement search and reload functionalities.
 - Handle loading states and API errors effectively.
-

3. Introduction

Modern web applications rely heavily on APIs to retrieve and exchange data with external services. Instead of storing all information locally, developers often use APIs to access real-time information from remote servers.

The **Random User Directory** project demonstrates this concept by using the **Random User API**, which generates realistic user data such as names, profile pictures, phone numbers, email addresses, and countries.

The application automatically fetches twenty random users when the webpage loads. Each user is displayed inside a responsive card containing their personal information. The project also provides a search feature that filters users instantly based on their names, making navigation easier. Additionally, a reload button allows users to request a completely new set of random users without refreshing the webpage.

This project strengthens the understanding of frontend development by combining HTML, CSS, and JavaScript to build an interactive and responsive web application.

4. Technologies Used

The following technologies were used to develop this project:

- **HTML5** – Used to create the structure of the webpage.
- **CSS3** – Used to design and style the application with a responsive layout.

- **JavaScript (ES6)** – Used to implement application logic and dynamic functionality.
- **Fetch API** – Used to retrieve data from the Random User API.
- **Visual Studio Code** – Used as the development environment.
- **Live Server Extension** – Used for testing the project locally.

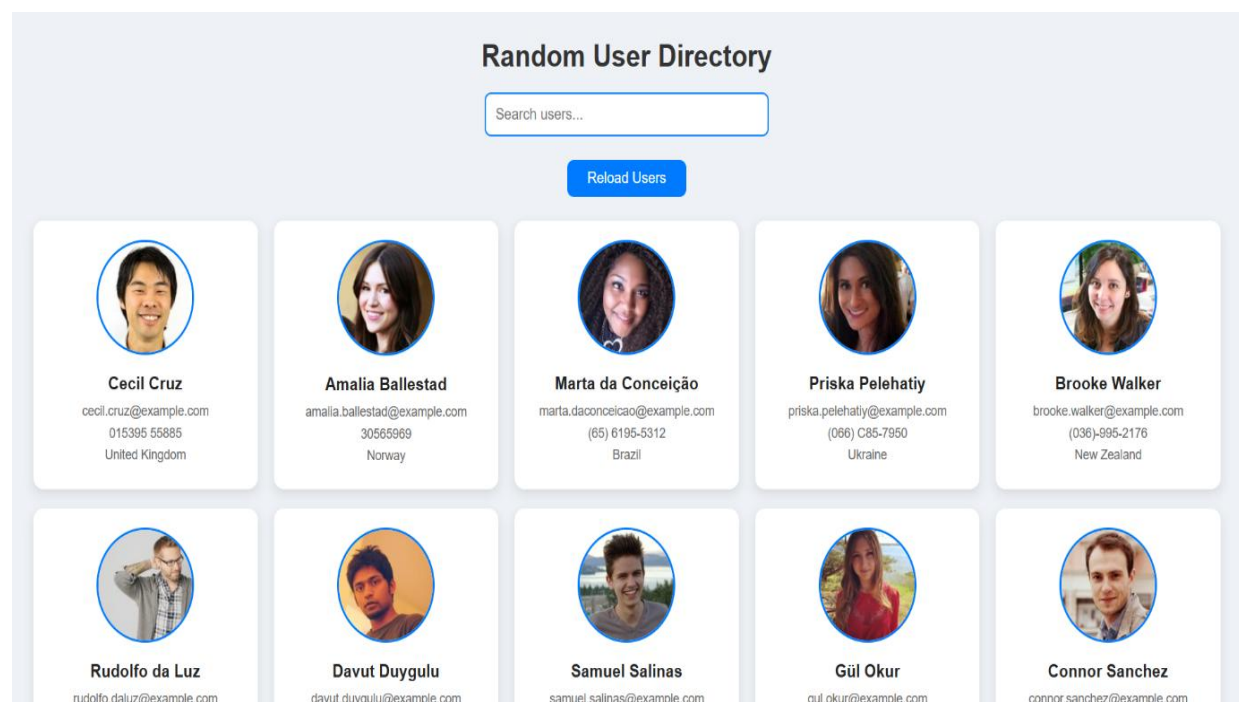
5. Project Description

The Random User Directory is a responsive web application that retrieves random user information through the Random User API.

When the webpage loads, JavaScript sends an asynchronous request to the API. After receiving the response, the user information is extracted and displayed dynamically as profile cards.

Each card contains:

- User Profile Picture
- Full Name
- Email Address
- Phone Number
- Country



To improve user experience, the application provides several interactive features. Users can search for a specific person using the search bar, which filters the displayed cards in real time. The reload button fetches a new list of random users instantly, while a loading indicator informs users that data is currently being retrieved. If the API request fails, an appropriate error message is displayed.

The responsive design ensures that the application works efficiently on desktops, tablets, and mobile devices.

6. Features

The application provides the following features:

- Integration with the Random User REST API.
- Dynamic retrieval of user information.
- Responsive card-based interface.
- Real-time search functionality.
- Reload button for fetching new users.
- Loading indicator during API requests.
- Error handling for failed API calls.
- Clean and user-friendly interface.
- Responsive design compatible with different screen sizes.

7. API Used

API Name

Random User API

API URL

<https://randomuser.me/api/>

```

{
  "results": [
    {
      "gender": "female",
      "name": {
        "title": "Ms",
        "first": "Abigail",
        "last": "Denys"
      },
      "location": {
        "street": {
          "number": 6416,
          "name": "Vimy St"
        },
        "city": "Westport",
        "state": "Saskatchewan",
        "country": "Canada",
        "postcode": "U6N 6N2",
        "coordinates": {
          "latitude": "-67.9362",
          "longitude": "103.6143"
        },
        "timezone": {
          "offset": "-3:30",
          "description": "Newfoundland"
        }
      },
      "email": "abigail.denys@example.com",
      "login": {
        "uuid": "04f5affe-e6ad-478c-bafd-6a38f33f0efd",
        "username": "ticklishgorilla714",
        "password": "zero",
        "salt": "AN18FUUQ",
        "md5": "84d709627ca64ac1869a1518a1729516",
        "sha1": "fbcd23219a0f990c590fa287176664532bc4dd5b",
        "sha256": "b30ab57b2ac9511bb00e21236e356b56a31328fd68e4970b3118273661195b04"
      },
      "dob": {
        "date": "1958-08-28T10:36:05.387Z",
        "age": 67
      },
      "registered": {
        "date": "2013-06-21T07:04:27.792Z",
        "age": 13
      },
      "phone": "F29 A99-8194",
      "cell": "N08 T32-9053",
      "id": {
        "name": "SIN",
        "value": "313750630"
      },
      "picture": {
        "large": "https://randomuser.me/api/portraits/women/95.jpg",
        "medium": "https://randomuser.me/api/portraits/med/women/95.jpg",
        "thumbnail": "https://randomuser.me/api/portraits/thumb/women/95.jpg"
      },
      "nat": "CA"
    }
  ],
  "info": {
    "seed": "3bfa2b2b26854b60",
    "results": 1,
    "page": 1,
    "version": "1.4"
  }
}

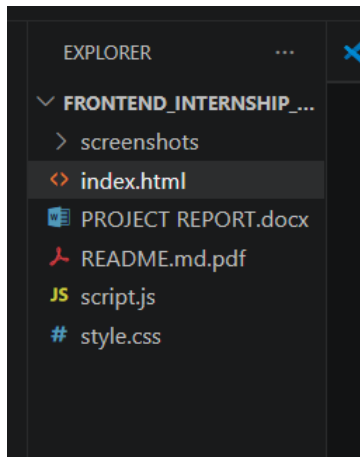
```

Purpose

The Random User API is a free public REST API that generates realistic user information. It provides data such as profile pictures, names, email addresses, phone numbers, addresses, gender, nationality, and other personal details.

For this project, twenty random users are fetched and displayed dynamically on the webpage.

8. Folder Structure



9. Methodology / Implementation

The project was developed using the following steps:

Step 1: Project Setup

A project folder was created in Visual Studio Code containing HTML, CSS, JavaScript, and README files.

Step 2: HTML Development

The webpage structure was designed using HTML elements such as headings, input fields, buttons, and containers.

Step 3: CSS Styling

CSS was used to create a modern, responsive, and visually appealing card layout with hover effects and proper spacing.

Step 4: API Integration

The Fetch API along with Async/Await was used to request user data from the Random User API.

Step 5: Dynamic Data Rendering

JavaScript processed the API response and dynamically generated user cards using DOM manipulation.

Step 6: Search Functionality

An event listener was implemented to filter displayed users based on the entered search text.

Step 7: Reload Functionality

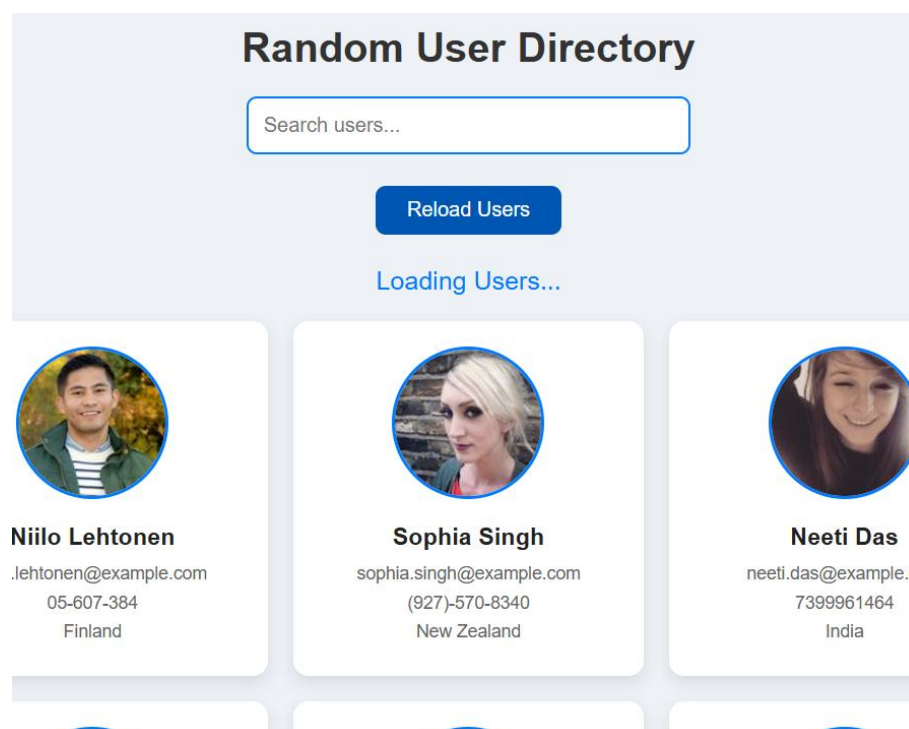
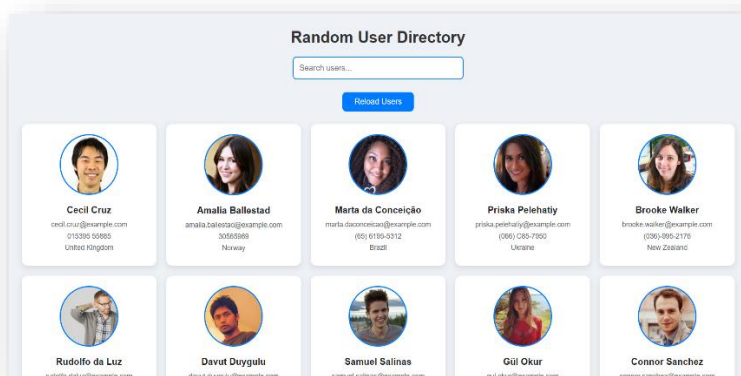
A reload button was added to fetch a completely new list of users without refreshing the webpage.

Step 8: Error Handling

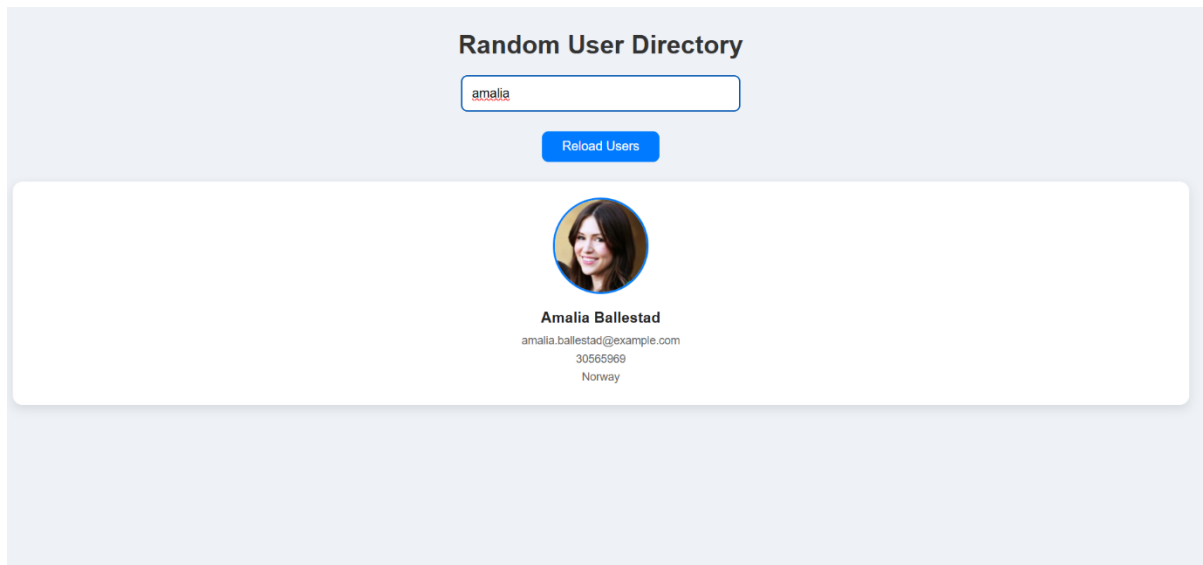
Try-Catch blocks were used to display an error message whenever the API request failed.

Step 9: Testing

The application was tested using the Live Server extension in Visual Studio Code to ensure proper functionality and responsiveness.



Reload



Searching users

10. Learning Outcomes

During the development of this project, the following concepts were learned:

- REST API Integration
- JavaScript Fetch API
- Asynchronous Programming using Async/ Await
- DOM Manipulation
- Event Handling
- Dynamic Content Rendering
- Responsive Web Design
- Error Handling Techniques
- Frontend Project Organization
- Debugging and Testing in Visual Studio Code

This project also improved problem-solving skills and provided practical experience in developing real-world frontend applications.

11. Conclusion

The **Random User Directory** project successfully demonstrates the practical implementation of frontend technologies with API integration. By fetching real-time data from the Random User API, the application provides a dynamic and interactive user experience.

The project effectively combines HTML, CSS, and JavaScript to create a responsive web application featuring user search, dynamic data rendering, loading indicators, reload functionality, and error handling. It serves as an excellent example of how frontend applications interact with external APIs to deliver meaningful content.

Overall, this project enhanced my understanding of modern web development practices, strengthened my JavaScript programming skills, and provided valuable hands-on experience in building responsive, API-driven web applications suitable for real-world use.

12. GitHub profile link

<https://github.com/nireekshamr>